NearSens - use and calibration guide

NearSens is a 100-500 mm distance meter prototype. It uses ultrasonic measurement and shows the result on a four-digit LCD.

This document describes startup, basic measurement, RAW mode, 75 mm correction, calibration and the codes shown by the device.

1. Power supply

Power the device from a 5 V DC supply. Connect power to the PS1 screw terminal on the PCB. Keep polarity according to the markings on the prototype enclosure.

After power-up, the display shows code 8888. This means device startup and readiness for measurement.

2. Test object preparation

For testing, use a flat, hard object placed as perpendicular as possible to the ultrasonic transducers.

The recommended object surface is about 20 x 20 cm or larger. This matches the example test object and gives a larger, more repeatable reflection surface for the ultrasonic wave.

For a normal test, set the distance from the front face of the ultrasonic transducers. In this setup, the 75 mm correction should be disabled.

Enable the 75 mm correction only when the result should be referenced to the rear wall of the enclosure, meaning the end of the device opposite to the transducers.

3. Basic measurement

1. Connect the 5 V DC supply.
2. Wait for code 8888.
3. Place the test object at the selected distance.
4. Briefly press TS3.
5. Read the result from the LCD.

The result is shown in millimetres. During measurement, the display may show a short digit animation. After measurement, either the result or an error code appears.

A computer, programmer or Arduino IDE is not needed for a normal test. A 5 V DC supply is enough.

4. Buttons

Element	Action
TS3	single measurement; stores the current point during calibration
TS4	toggles 75 mm correction; selects the calibration point during calibration
TS3 + TS4	enters calibration after holding for about 5 s
5 quick TS3 clicks	enables or disables RAW mode

5. 75 mm reference correction

The 75 mm correction changes only the value shown on the LCD. It does not change calibration points stored in EEPROM and does not apply in RAW mode.

In BODY mode, the firmware subtracts 75 mm from the normal result. This only makes sense when the measurement should be referenced to the rear wall of the enclosure, not to the front of the transducers.

Codes shown after pressing TS4:

Code	Meaning
0 _ 75	75 mm correction disabled
1 _ 75	75 mm correction enabled

For measurement from the transducer front face, use mode 0 _ 75.

6. RAW mode

RAW mode shows the raw result before calibration correction and before the 75 mm reference correction. It is used for measurement path diagnostics.

To toggle RAW, quickly press TS3 five times.

Code	Meaning
7001	RAW mode enabled
7000	RAW mode disabled

RAW should be disabled for normal measurement. RAW is only for diagnostics.

7. Calibration

Calibration stores reference points in the microcontroller EEPROM. Recommended calibration points are:

- 100 mm
- 200 mm
- 300 mm
- 400 mm
- 500 mm

Procedure:

1. Hold TS3 and TS4 together for about 5 s.
2. Select the calibration point with TS4.
3. Place the object at the selected distance.
4. Press TS3 to store the point.
5. Repeat for the next points.

Code 1111 means that the calibration point was stored correctly. If an error code appears, check object placement, distance and echo stability, then repeat the point storage.

Calibration should be performed in the same setup in which later measurements will be made. The prototype does not have an air temperature sensor, so temperature changes can affect the result by changing the speed of sound.

8. LCD codes

Code	Meaning
8888	device startup and ready for measurement
1111	calibration point stored correctly
7001	RAW mode enabled
7000	RAW mode disabled
0 _ 75	75 mm correction disabled
1 _ 75	75 mm correction enabled
4000	general no-result code
4001	no stable echo
4002	echo too early, crosstalk or transmitter blanking
4003	echo outside technical range
4004	calibration error
4005	no valid calibration data
4006	calibration sample spread too large
4007	ADC/RX receiver path error

9. If the result is unstable

- check that the object is flat and perpendicular to the transducers
- increase the test object surface area
- avoid soft or sound-absorbing materials
- limit reflections from nearby objects
- check that the distance is within 100-500 mm
- calibrate at 100, 200, 300, 400 and 500 mm
- if needed, enable RAW and compare the raw result with the calibrated result